UNIT 2 · TOPIC 4(B)

Stability of Farm Machines

Syllabus topic: Dynamic balancing and stability of farm machines

Static stability on slopes

A machine stays statically stable while the vertical through its centre of gravity (CG) falls inside the wheel-contact polygon.

$$\text{Lateral overturn limit: } \tan\theta = (\text{wheel track}/2) \div \text{CG height} = B/(2h)$$

$$\text{Longitudinal (uphill) limit: } \tan\theta = \text{horizontal distance of CG from rear axle} \div \text{CG height}$$

- A wider track and a lower CG raise the lateral limit.
- Hillside combines use body-levelling systems; test codes specify slope limits for safe operation of self-propelled combines and trailers.

Mounted/trailed implements and weight transfer

- Raising a heavy rear-mounted implement moves the tractor–implement CG rearward and can lift the front. Keep a minimum of **~20%** of total weight on the front axle (steering control); add front ballast per a moment balance about the rear axle.
- A trailed implement pulling below/above the line of pull changes the vertical hitch-point loads. A high hitch transfers weight to the tractor rear wheels (traction gain but front-end lightness). The hitch geometry must keep the resultant pull through a safe zone (full force analysis — Topic 5, and Unit 4 hitching).
- **Transport stability:** mounted implements carried high raise the CG; folding wide implements reduces transport width and lateral moment; drawbar trailers need a positive tongue load (**~10–15% of gross**) to prevent snaking.

Dynamic stability

- At speed, ground undulations cause bouncing/pitching — trailed-machine wheels and tyres act as a spring-mass system. Check that gang bounce (disc harrows) does not leave unworked strips.
- Damp snaking/weaving of trailed machines at transport speed with proper tongue load and hitch length.
- PTO-driven machines: driveline angularity during turns causes torsional oscillation — limit joint angles; use CV joints for wide-angle turning.

Practice MCQs

Q66. A machine on a side slope overturns when the vertical through the CG:

(a) Passes inside wheelbase
(b) Falls outside the wheel-contact polygon
(c) Is at axle height
(d) Coincides with draft line

Q67. Lateral overturn limit angle with track width B and CG height h is given by:

(a) tan θ = 2h/B
(b) tan θ = B/2h
(c) sin θ = B/h
(d) θ = B·h

Q68. Lateral stability is improved by:

(a) Higher CG, narrow track
(b) Lower CG, wider track
(c) More speed
(d) Higher hitch

Q69. With a heavy rear-mounted implement raised, safe steering requires the front axle to retain at least about:

(a) 5%
(b) 20%
(c) 40%
(d) 60% of total weight

Q70. Front-end ballast requirement for a mounted implement is computed by:

(a) Energy balance
(b) Moment balance about the rear axle
(c) Draft equation
(d) Traction equation

Q71. Positive tongue (hitch) load recommended on trailed trailers is about:

(a) 0%
(b) 10–15% of gross weight
(c) 30–40%
(d) 50%

Q72. Snaking/weaving of a trailed implement at transport speed is reduced by:

(a) Negative tongue load
(b) Adequate tongue load and proper hitch length
(c) Higher speed
(d) Wider tyres

Answers

Q66 — (b) Falls outside the wheel-contact polygon
Q67 — (b) tan θ = B/2h — *At tipping, the CG vertical passes through the downhill wheel contact: tan θ = (B/2)/h.*
Q68 — (b) Lower CG, wider track
Q69 — (b) 20%
Q70 — (b) Moment balance about the rear axle
Q71 — (b) 10–15% of gross weight
Q72 — (b) Adequate tongue load and proper hitch length

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 1 (Research & Development), Ch. 2 (Field Capacities), §14.11 (Reciprocating Unbalance in a Mower); theory-of-machines standards (balancing, flywheels); tractor–implement mechanics references.